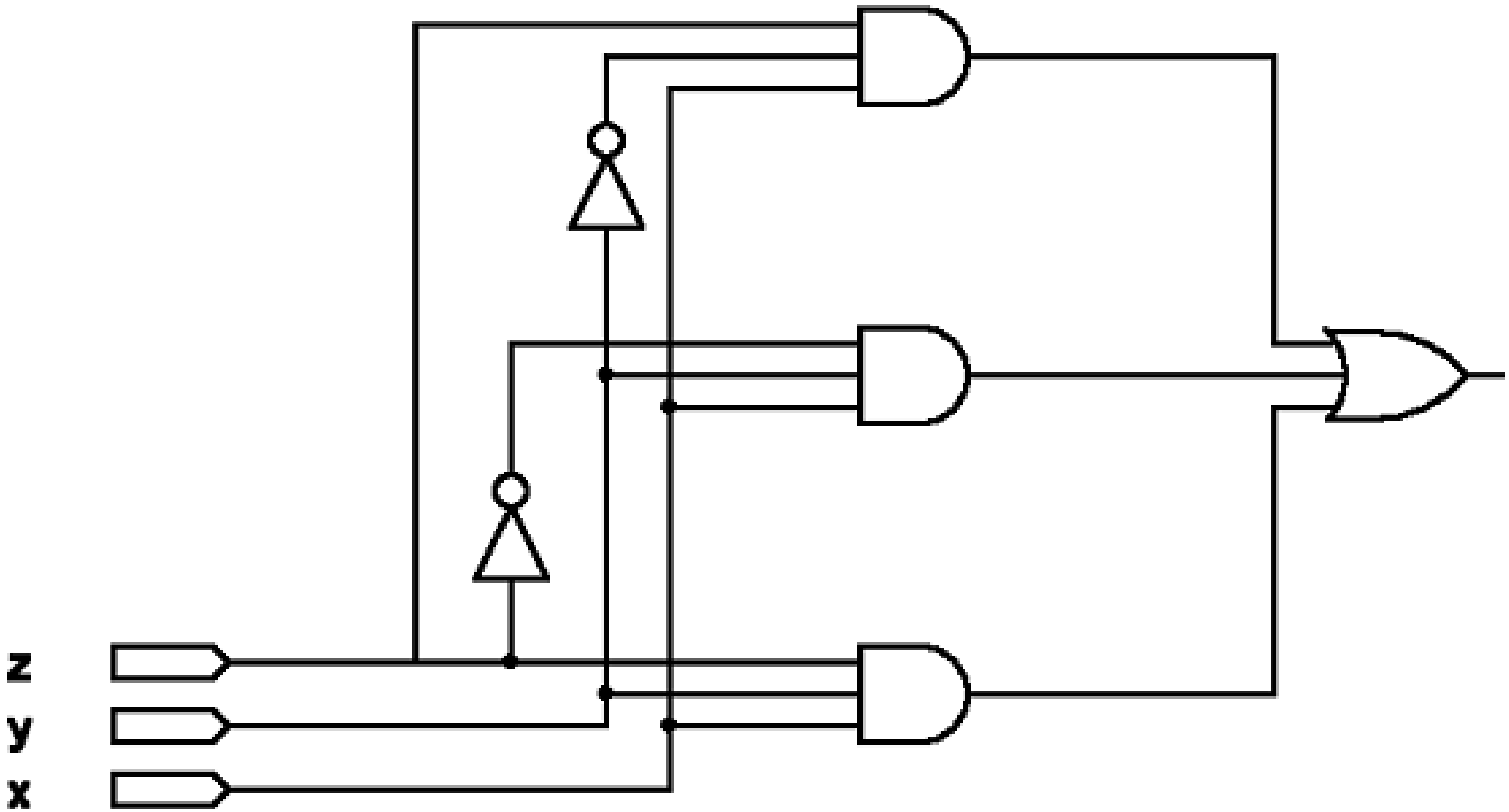
Introduction to Logic Circuits:

Synthesis using AND, OR, and NOT gates

Example logic circuit design

- Assume we want to design a logic circuit with three inputs *x*, *y*, and *z*
- The circuit output should be 1 only when *x*=1 and either *y* or *z* (or both) is 1
  - Three possible combinations
    - x=1, y=0, z=1 => $xy'z$
    - x=1, y=1, z=0 => xyz'
    - x=1, y=1, z=1 => xyz
- The function could be written as
  - $f(x,y,z)=xy'z+xyz'+xyz$
    - sum of products form

Example logic circuit design



Example logic circuit design

- Implements f correctly, $f(x,y,z)=xy'z+xyz'+xyz$ but is not the simplest such network

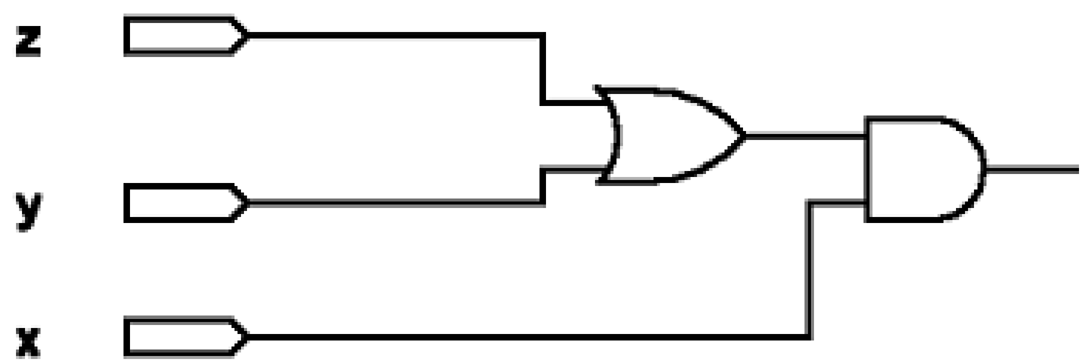
x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$xy'z+xyz'+xyz$

$xy'z+xy$ from 14a

$x(y'z+y)$ from 12a

$x(y+z)$ from 16a



Example logic circuit design

- Obviously, the cost (in terms of gates and connections) of this network is much less than the initial network
- The process of generating a circuit from a stated desired functional behavior is called ***synthesis***
- Generation of AND-OR style networks from a truth table is one of many types of synthesis techniques that we will cover

Logic synthesis

- If a function *f* is described in a truth table, then an expression that generates *f* can by obtained (synthesized) by
  - Considering all rows in the table where *f*=1, or
  - By considering all rows in the table where *f*=0
- This will be an application of the principal of duality

Minterms

- For a function of *n* variables f(a,b,c,...n)
  - A minterm of *f* is a product of *n* literals (variables) in which each variable appears once in either true or complemented form, but not both
    - f(a,b,c) -- minterm examples: abc, a'bc, abc'
    - f(a,b,c) -- invalid examples: ab, c', a'c
  - An *n* variable function has 2^n valid minterms

Minterms

Row number	*x*	*y*	*z*	*Minterm*
0	0	0	0	$m_0 = x'y'z'$
1	0	0	1	$m_1 = x'y'z$
2	0	1	0	$m_2 = x'yz'$
3	0	1	1	$m_3 = x'yz$
4	1	0	0	$m_4 = xy'z'$
5	1	0	1	$m_5 = xy'z$
6	1	1	0	$m_6 = xyz'$
7	1	1	1	$m_7 = xyz$

- Each row of a truth row corresponds to a single minterm
- When a function is written as a sum of minterms, the form is called a standard (or canonical) ***sum-of-products***

Minterm notation

- An equation may be written in terms of m-notation

$f(a,b,c)=m_0+m_1+m_2+m_4$

$f(a,b,c)=a'b'c'+a'b'c+a'bc'+ab'c'$

$$\underbrace{000}_{0} \quad \underbrace{001}_{1} \quad \underbrace{010}_{2} \quad \underbrace{100}_{4}$$

$f(a,b,c)=\Sigma m(0,1,2,4)$

a	*b*	*c*	*f*
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

Minterm notation examples

- What is the minterm notation for the following function?
  - $f(a,b,c)=abc+a'bc+abc'+a'b'c$

- What is the function (in terms of variables) if the minterm notation is the following?
  - $f(a,b,c)= \Sigma m(1,5,6,7)$

Logic synthesis

- Duality suggests that:
  - If it is possible to synthesize a function f by considering the truth table rows where f=1, then it should also be possible to synthesize f by considering the rows for which f=0.
- This approach uses the complement of minterms, which are called ***maxterms***

Maxterms

Row number	*x*	*y*	*z*	*Maxterm*
0	0	0	0	$M_0=x+y+z$
1	0	0	1	$M_1=x+y+z'$
2	0	1	0	$M_2=x+y'+z$
3	0	1	1	$M_3=x+y'+z'$
4	1	0	0	$M_4=x'+y+z$
5	1	0	1	$M_5=x'+y+z'$
6	1	1	0	$M_6=x'+y'+z$
7	1	1	1	$M_7=x'+y'+z'$

- Each row of a truth row corresponds to a single maxterm
- When a function is written as a product of maxterms, the form is called a standard (or canonical) ***product-of-sums***

Maxterm notation

- An equation may be written in terms of M-notation

$f(a,b,c)=M_3 \cdot M_5 \cdot M_6 \cdot M_7$

$f(a,b,c)=(a+b'+c')(a'+b+c')(a'+b'+c)(a'+b'+c')$

$$\underbrace{0\ 1\ 1}_{3} \quad \underbrace{1\ 0\ 1}_{5} \quad \underbrace{1\ 1\ 0}_{6} \quad \underbrace{1\ 1\ 1}_{7}$$

$f(a,b,c)=\Pi M(3,5,6,7)$

a	*b*	*c*	*f*
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

Maxterm notation examples

- What is the maxterm notation for the following function?
  - $f(a,b,c)=(a+b+c)(a'+b+c)(a+b+c')(a'+b'+c)$

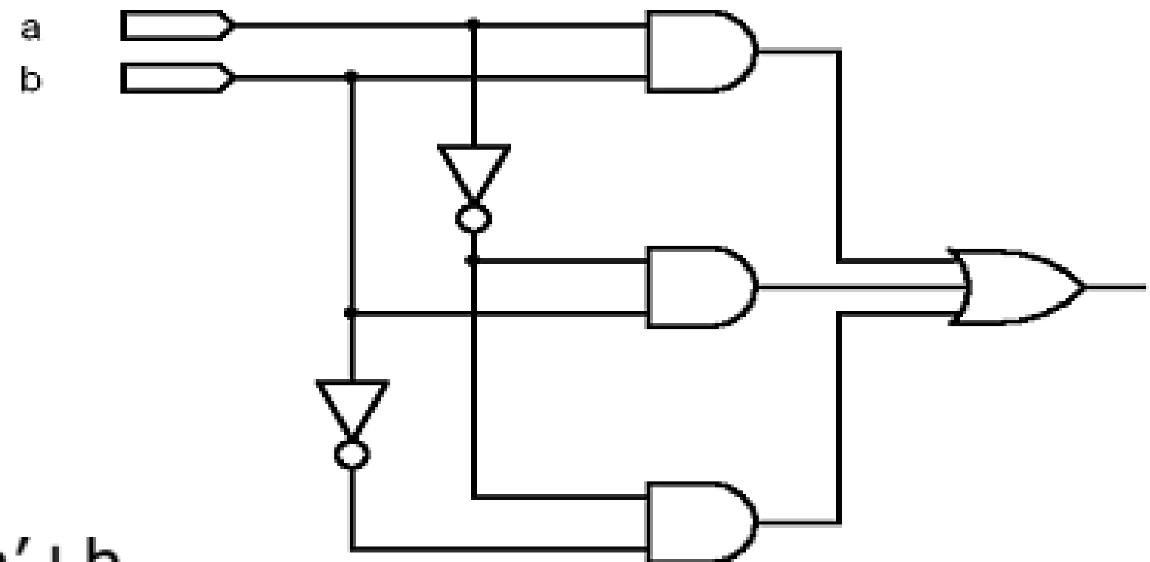
- What is the function (in terms of variables) if the maxterm notation is the following?
  - $f(a,b,c)= \Pi M(1,5,6,7)$

Sum-of-products and minimality

- A function expressed in standard sum-of-products (or product-of-sums) form may not be minimal

a	*b*	*f*
0	0	1
0	1	1
1	0	0
1	1	1

$f(a,b)=a'b'+a'b+ab=a'+b$



Design examples

- Logic circuits provide a solution to a problem
- Some may be complex and difficult to design
- Regardless of the complexity, the same basic design issues must be addressed
  1. Specify the desired behavior of the circuit
  2. Synthesize and implement the circuit
  3. Test and verify the circuit

Three-way light control

- Assume a room has three doors and a switch by each door controls a single light in the room.
  - Let x, y, and z denote the state of the switches
  - Assume the light is off if all switches are open
  - Closing any switch turns the light on. Closing another switch will have to turn the light off.
  - Light is on if any one switch is closed and off if two (or no) switches are closed.
  - Light is on if all three switches are closed

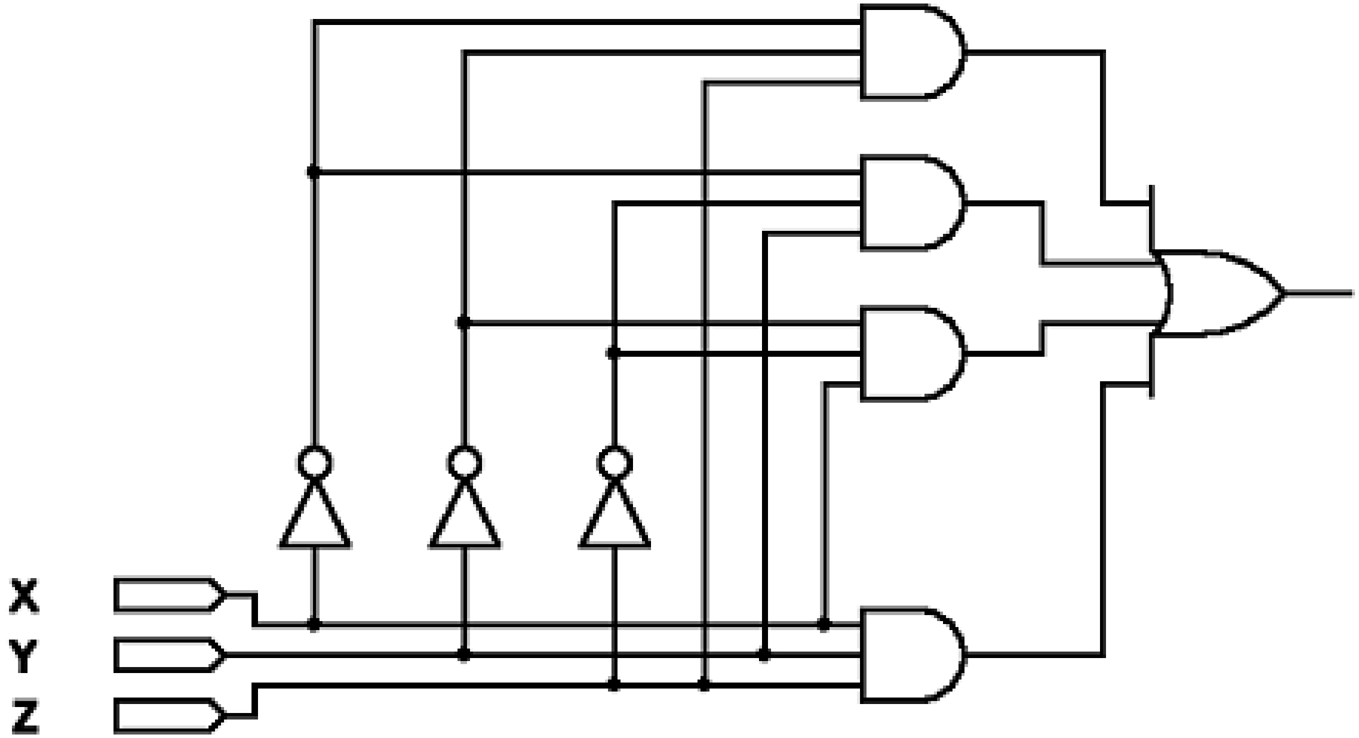
Three-way light control

x	y	z	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$f(x,y,z)=m_1+m_2+m_4+m_7$$

$$f(x,y,z)=x'y'z+x'yz'+xy'z'+xyz$$

This is the simplest sum-of-products form.



Multiplexer circuit

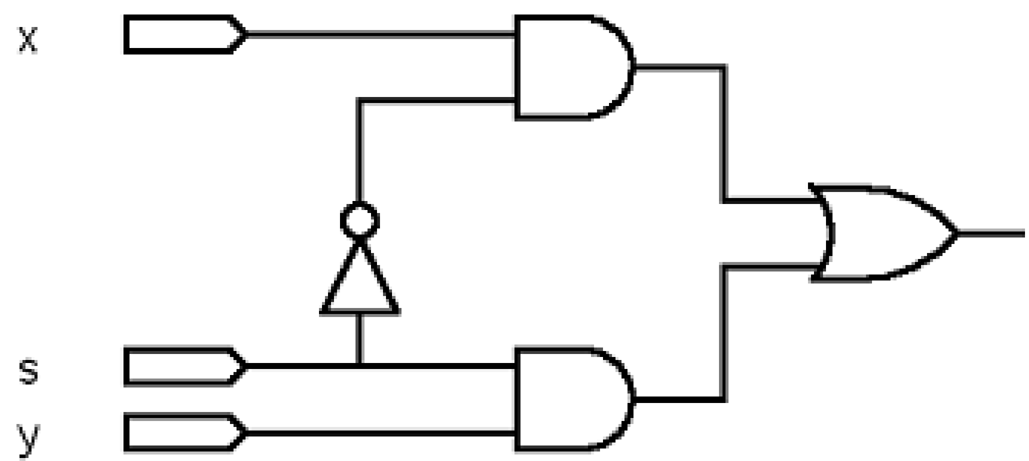
- In computer systems it is often necessary to choose data from exactly one of a number of sources
  - Design a circuit that has an output (f) that is exactly the same as one of two data inputs (x,y) based on the value of a control input (s)
    - If s=0 then f=x
    - If s=1 then f=y
  - The function f is really a function of three variables (s,x,y)
  - Describe the function in a three variable truth table

Multiplexer circuit

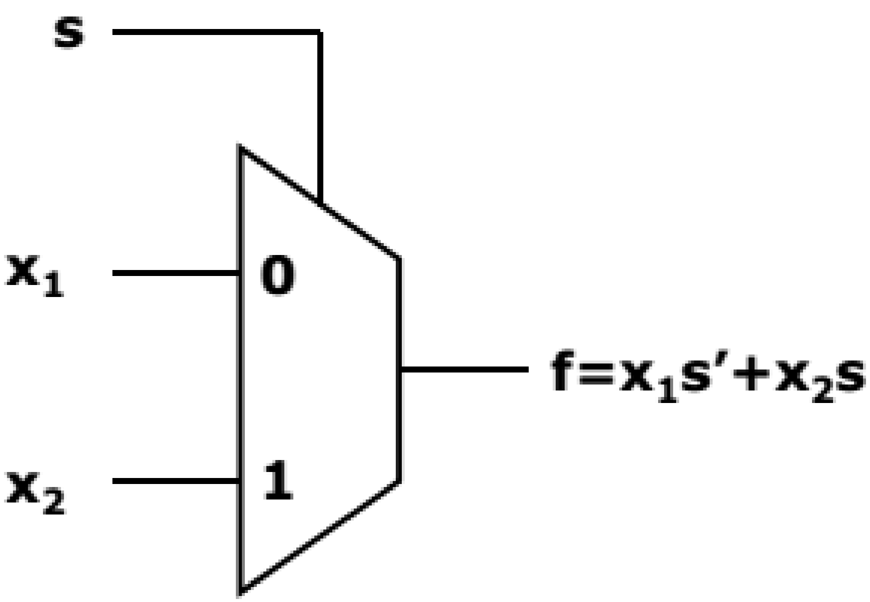
s	*x*	*y*	*f*
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

convenient to put control signal on left

$$f(s,x,y)=m_2+m_3+m_5+m_7$$
$$f(s,x,y)=s'xy'+s'xy+sx'y+sxy$$
$$f(s,x,y)=s'x(y'+y)+sy(x'+x)$$
$$f(s,x,y)=s'x+sy$$



Multiplexer circuit



Graphical symbol

s	$f(s,x_1,x_2)$
0	x_1
1	x_2

Compact truth table

Car safety alarm

- Design a car safety alarm considering four inputs
  - Door closed (D)
  - Key in (K)
  - Seat pressure (S)
  - Seat belt closed (B)
- The alarm (A) should sound if
  - The key is in and the door is not closed, or
  - The door is closed and the key is in and the driver is in the seat and the seat belt is not closed

Car safety alarm

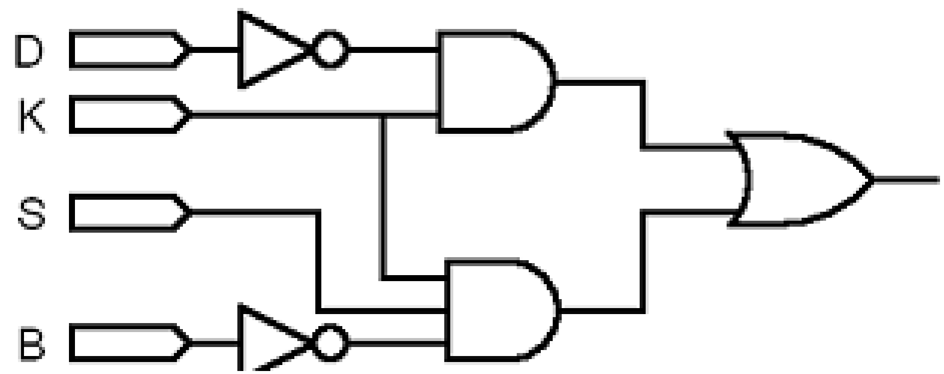
D	K	S	B	A
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

$A(D,K,S,B)=\Sigma m(4,5,6,7,14)$

$A(D,K,S,B)=D'KS'B'+D'KS'B+D'KSB'+D'KSB+DKSB'$

$=D'KS'+D'KS+KSB'$

$=D'K+KSB'$



Majority circuit

- Design a circuit with three inputs (x,y,z) whose output (f) is 1 only if a majority of the inputs are 1
  - Construct a truth table
  - Write a standard sum-of-products expression for f
  - Draw a circuit diagram for the sum-of-products expression
  - Minimize the function using algebraic manipulation
    - During your minimization you can use any Boolean theorem, but leave the result in sum-of-products form (generate a minimum sum-of-products expression)
  - Draw the minimized circuit